ATARS v2.0

Automated Time-Series Analysis and Reporting System

A Formal Statistical Framework with Machine Learning Enhancements and Algorithmic LLM Grounding Verification for Urban Air Quality Analysis

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Abstract

Urban air quality monitoring stations collectively generate tens of millions of sensor records per year, yet no open, reproducible, equation-numbered analysis pipeline exists that integrates data quality scoring, formal statistical operators, machine learning anomaly detection, short-range forecasting, and automated report generation into a single auditable system. This paper presents **ATARS v2.0** (Automated Time-Series Analysis and Reporting System), which addresses this gap through three interleaved contributions implemented as a single 4,449-line open-source Python module.

ATARS v2.0 defines **16 formally numbered core statistical operators** (Eqs. 1–16) covering aggregation A(D_v), quality scoring Q(D), rolling baseline B(D,W), z-score anomaly detection ζ_d, delta deviation δ_d, confidence intervals CI_{d,v}, autocorrelation (ACF/PACF), OLS regression, Pearson correlation, and the novel grounding constraint G(s) ⊆ J. These are supplemented by equations for the ML modules (Eqs. 18–23), the grounding verifier algorithm (Eqs. 24–26), and the SHA-256 reproducibility audit (Eqs. 27–28), for a total of 28 numbered equations.

Two additive machine learning modules — Isolation Forest multivariate anomaly detection and Holt-Winters triple exponential smoothing forecasting — extend the system's analytical capabilities without modifying any formal operator. The primary novel contribution is the **Runtime Grounding Verifier (RGV)**, which introduces the first known algorithmic enforcement and measurement of the constraint G(s) ⊆ J for LLM-generated narrative in an environmental monitoring pipeline. The scalar metric G_rate ∈ [0, 1] quantifies numerical grounding with type-adaptive tolerance bounds.

The system was validated against synthetic datasets with analytically known ground truth, and demonstrated on a **real-world case study: Gurugram, Haryana, India (January–December 2024)**, using publicly available CPCB air quality data (8,784 hourly records, 10 pollutant variables). The real-world demonstration achieved G_rate = 92.9% (PASS), identified 19 Isolation Forest anomaly days (16 multivariate-only), and produced an in-sample Holt-Winters RMSE of 77.89 μg/m³ for PM10. The full pipeline completed in under 5 minutes.

Keywords: air quality analysis · time-series automation · Isolation Forest · Holt-Winters exponential smoothing · LLM grounding verification · G_rate metric · runtime verifier · reproducibility · formal statistical operators · CPCB · Gurugram

1. Introduction

Urban air pollution is one of the most consequential public health crises of the twenty-first century. The World Health Organization estimates that ambient air pollution causes approximately 4.2 million premature deaths globally each year [1]. Ground-level monitoring stations continuously record concentrations of particulate matter (PM10, PM2.5), nitrogen oxides (NO, NO2, NOx), sulphur dioxide (SO2), carbon monoxide (CO), ozone, ammonia, and volatile organic compounds (VOCs) at hourly intervals. A single station operating continuously generates approximately 61,320–87,648 records per year. Nationally operated networks — such as India's CPCB Continuous Ambient Air Quality Monitoring System (CAAQMS), which covers over 800 stations across 344 cities — produce hundreds of millions of records annually.

Despite the volume and public health importance of this data, a systematic survey of published urban air quality studies reveals a consistent reproducibility failure: analysis pipelines are rarely documented with equation-level precision, almost never publicly released with runnable code, and essentially never accompanied by a formal audit trail linking computed statistics to generated report text. Existing tools such as openair [3] provide excellent exploratory analysis capabilities but do not integrate automated report generation, formal anomaly detection with z-score operators, or machine learning enhancement into a single reproducible pipeline.

The recent integration of Large Language Models (LLMs) into data-driven reporting introduces a second, qualitatively distinct failure mode: hallucination. An LLM tasked with writing a narrative about sensor measurements may confidently produce numerical claims — exceedance counts, correlation coefficients, anomaly dates — that were never present in the input data. In an environmental monitoring or regulatory reporting context, such fabricated values could influence policy decisions, trigger unnecessary alerts, or undermine public trust in monitoring systems. Prior systems addressed this risk only through prompt instruction — a probabilistic, unauditable mechanism that cannot be independently verified.

This paper presents ATARS v2.0, which addresses both failure modes simultaneously through three interleaved contributions: (1) a complete, equation-numbered, formally verifiable statistical analysis pipeline for urban air quality data; (2) two additive machine learning modules that expand detection and forecasting capabilities; and (3) the Runtime Grounding Verifier (RGV) — the primary novel contribution — which algorithmically enforces and measures the grounding constraint $G(s) \subseteq J$ (Eq. 15) for every numerical claim in LLM-generated narrative text.

1.1 Paper Organisation

Section 2 surveys related work across air quality frameworks, ML anomaly detection and forecasting, and LLM grounding. Section 3 defines the 16 core formal operators. Section 4 describes the 9-step

system architecture. Sections 5 and 6 detail the ML modules and the RGV. Section 7 covers the security model. Section 8 presents validation — both synthetic and a real-world case study on Gurugram 2024 CPCB data. Section 9 provides an honest novelty assessment. Section 10 concludes.

2. Related Work

2.1 Air Quality Analysis Frameworks

Published air quality analysis tools broadly partition into three categories: regulatory compliance systems, research packages, and visualisation dashboards. Regulatory systems such as the US EPA Air Quality System (AQS) and India's CPCB CAAQMS compute threshold exceedances and produce standardised reports, but neither defines formal mathematical operators, computes rolling baseline anomalies, nor supports automated narrative generation [2]. The openair R package [3] provides a rich suite of exploratory analysis functions — including polar plots, trajectory analysis, and time-variation plots — and represents the closest prior art to ATARS in terms of scientific scope. However, openair does not define equation-numbered formal operators, does not generate automated Word reports, and contains no LLM integration or grounding verification.

Dashboard platforms such as IQAir and AirVisual display real-time AQI values but are not designed for retrospective time-series analysis, formal anomaly detection, or automated report generation [4]. None of these systems defines or measures a grounding constraint for any AI-generated content. ATARS addresses this gap by providing a complete, formally specified, end-to-end pipeline from raw CSV ingestion to a verified, auditable Word document report.

2.2 Anomaly Detection in Environmental Time Series

Classical univariate anomaly detection for environmental data relies predominantly on z-score thresholding (Gauss, 1809) [5]. Hampel identifiers and IQR-based filters are also widely used but share the z-score's fundamental limitation: they evaluate each variable independently and cannot detect anomalous joint behaviour across multiple pollutants. This univariate constraint means that a day on which PM10, NO2, and SO2 are simultaneously elevated by 2.5σ each — a pattern strongly indicative of a regional pollution episode — would not be flagged by any single z-score check.

Liu et al. (2008) introduced the Isolation Forest algorithm, which identifies anomalies by measuring how few partitions are required to isolate a data point in a randomly partitioned sub-space [6]. Unlike density-based methods (LOF, DBSCAN), Isolation Forest requires no distributional assumption, is linear in complexity, and handles correlated features naturally — making it well-suited for the high-dimensional daily pollutant matrices ATARS constructs. LSTM-based anomaly detection [7] and Support Vector Data Description [8] have also been applied to environmental time series but require substantially more training data and introduce hyperparameters that complicate reproducibility auditing.

2.3 Forecasting in Environmental Monitoring

Short-range (1–14 day) air quality forecasting spans classical exponential smoothing to deep learning. Holt-Winters triple exponential smoothing (Winters, 1960) [9] captures level, trend, and additive or multiplicative seasonality in a single interpretable model with three parameters (α, β, γ). Its transparency makes it particularly appropriate for a system where reproducibility and auditability are primary requirements, as all parameters are logged in the SHA-256 audit record.

Neural approaches including temporal convolutional networks [10], transformer-based models [11], and Prophet [12] demonstrate superior long-sequence accuracy but introduce numerous hyperparameters that complicate reproducibility auditing, require substantially more training data, and produce forecasts that are difficult to interpret without domain expertise. ATARS deliberately uses Holt-Winters rather than neural approaches to prioritise interpretability and reproducibility over maximum forecast accuracy.

2.4 LLM Grounding and Hallucination Mitigation

Hallucination in large language models — the confident generation of factually incorrect text — has been extensively studied since the GPT-3 era [13]. Mitigation strategies include retrieval-augmented generation (RAG) [14], chain-of-thought prompting [15], constrained decoding [16], and post-hoc fact verification [17]. These approaches are designed for general knowledge grounding and operate over open-domain text or web-retrieved documents.

FactScore [18] and FaithDial [19] provide sentence-level faithfulness metrics for LLM outputs. However, both are designed for natural language entailment rather than numerical value verification: FactScore decomposes claims into atomic facts and verifies each via web search retrieval, while FaithDial measures faithfulness in information-seeking dialogue. Neither system operates on a closed, typed numerical data contract. Critically, FactScore requires an external retrieval model (typically a web search engine or dense retrieval index) that is unavailable in a local, zero-transmission environment; FaithDial requires a labelled dialogue corpus. By contrast, RGV requires only J — a JSON object already produced as a standard pipeline artifact — and executes in seconds using pure Python. RLHF-based approaches [20] reduce hallucination at training time but provide no per-run verification artifact that can be included in a reproducibility audit log.

Positioning of RGV: The RGV is distinct from all prior work in three specific ways: (1) it verifies numerical values — not semantic meaning or factual propositions — against a typed, closed data contract J; (2) it uses type-adaptive tolerance bounds calibrated to the measurement characteristics of environmental data (±5% for concentrations, ±2 for integer counts, ±0.1 for quality scores); and (3) it is embedded as a runtime pipeline step producing a machine-readable audit artifact (grounding_verification.json) alongside the report. To the best of the author's knowledge, no prior published system combines these three properties.

3. Formal Mathematical Framework

Let V = {PM10, NO2, SO2, CO, Ozone, Benzene, Humidity, ...} be the set of monitored variables, T = {t1, t2, ..., tn} the set of hourly timestamps, and D = {(t, v, x_{t,v}, q_{t,v})} the complete dataset where x_{t,v} ∈ ℝ is the sensor reading and q_{t,v} ∈ {0,1} the quality flag. All 16 core operators are applied per-variable and per-day unless otherwise specified. All operators are pure deterministic functions of their inputs — a property that underpins the Reproducibility Theorem (Eq. 17).

3.1 Quality Operators

3.1.1 Quality Flag Assignment (Eq. 1)

Each record is quality-flagged based on physical plausibility bounds [l_v, u_v] specific to variable v:

$$q_{t,v} = 1 \text{ if } l_v \le x_{t,v} \le u_v, \text{ else } 0 \quad \textit{(Eq. 1)}$$

Records with q_{t,v} = 0 are excluded from all downstream computation. Physical bounds: PM10 ∈ [0, 1000] μg/m³; NO2, SO2 ∈ [0, 500] μg/m³; CO ∈ [0, 50] mg/m³; Ozone ∈ [0, 400] μg/m³; Benzene ∈ [0, 100] μg/m³; Humidity ∈ [0, 100]%.

3.1.2 Quality Score Q(D) (Eq. 2)

The quality score for variable v measures data completeness as the ratio of valid to expected records:

$$Q(D_v) = N_v / N_{total} \text{ where } N_v = |\{t \in T : q_{t,v} = 1\}|, \; N_{total} = 24 \times |days| \quad \textit{(Eq. 2)}$$

The overall dataset quality Q(D) = min_v Q(D_v) provides a conservative bound. Confidence classification: HIGH if Q ≥ 0.90; MODERATE if 0.70 ≤ Q < 0.90; LOW if Q < 0.70.

3.2 Aggregation Operator A(D_v) (Eqs. 3–5)

For each day d and variable v, the daily aggregation operator computes:

$$\bar{x}_{d,v} = (1/N_{d,v}) \times \Sigma_{t \in d,\, q=1} x_{t,v} \quad \textit{(Eq. 3)}$$

$$\sigma_{d,v} = \sqrt{[(1/(N_{d,v}-1)) \times \Sigma_{t \in d}(x_{t,v} - \bar{x}_{d,v})^2]} \quad \textit{(Eq. 4)}$$

$$N_{d,v} = |\{t \in d : q_{t,v} = 1\}| \quad \textit{(Eq. 5)}$$

Days with N_{d,v} = 0 (complete data loss) are excluded from baseline and anomaly calculations.

3.3 Baseline Operators B(D, W) (Eqs. 6–7)

The rolling baseline operator uses a centred window of W = 30 days. Centering prevents lookback bias at the cost of requiring future values — acceptable for retrospective batch analysis:

$$\bar{x}_{W,d} = (1/|W_d|) \times \Sigma_{d' \in W_d} \bar{x}_{d',v} \quad \text{where} \quad W_d = \{d' : |d'-d| \leq W/2,\ d' \neq d\} \qquad \textit{(Eq. 6)}$$

$$\sigma_{W,d} = \sqrt{[(1/(|W_d|-1)) \times \Sigma_{d' \in W_d}(\bar{x}_{d',v} - \bar{x}_{W,d})^2]} \qquad \textit{(Eq. 7)}$$

3.4 Anomaly Detection Operators (Eqs. 8–9)

3.4.1 Z-Score Operator ζ_d (Eq. 8)

The z-score measures daily deviation from the local baseline in units of baseline standard deviation:

$$\zeta_{d,v} = (\bar{x}_{d,v} - \bar{x}_{W,d}) / \sigma_{W,d} \qquad \textit{(Eq. 8)}$$

Day d is flagged as anomalous if $|\zeta_{d,v}| > \zeta_{thresh} = 3.0$ (default). Under a Normal distribution, $|\zeta| > 3.0$ has probability ≈ 0.27%, making this a conservative threshold appropriate for rare pollution episodes.

3.4.2 Delta Percentage Deviation δ_d (Eq. 9)

An interpretable complement to the z-score expressed as percent deviation from baseline:

$$\delta_{d,v} = [(\bar{x}_{d,v} - \bar{x}_{W,d}) / \bar{x}_{W,d}] \times 100\% \qquad \textit{(Eq. 9)}$$

3.5 Confidence Interval Operator (Eq. 10)

The 95% confidence interval of each daily mean under the Central Limit Theorem:

$$CI_{d,v} = \bar{x}_{d,v} \pm 1.96 \times \sigma_{d,v} / \sqrt{N_{d,v}} \qquad \textit{(Eq. 10)}$$

CI width is inversely proportional to $\sqrt{N_{d,v}}$: days with fewer valid records produce wider intervals, directly reflecting reduced precision.

3.6 Temporal Dependency Operators (Eqs. 11–12)

3.6.1 Autocorrelation Function (Eq. 11)

$$\hat{\rho}(k) = \Sigma_{t=k+1}^{n} (x_t - \bar{x})(x_{t-k} - \bar{x}) / \Sigma_{t=1}^{n} (x_t - \bar{x})^2 \qquad \textit{(Eq. 11)}$$

Significant $\hat{\rho}(k)$ at lag 7 indicates weekly seasonality. The 95% significance band is $\pm 1.96/\sqrt{n}$.

3.6.2 Partial Autocorrelation Function (Eq. 12)

$$\Phi_k = R_k^{-1} r_k \quad \text{where} \quad R_k \text{ is the lag-k autocorrelation matrix, } r_k = [\hat{\rho}(1),...,\hat{\rho}(k)]^T \qquad \textit{(Eq. 12)}$$

PACF cutoff at lag p indicates an AR(p) process. Jointly, ACF decay and PACF cutoff identify ARMA(p,q) model orders.

3.7 Multivariate Operators (Eqs. 13–14)

3.7.1 Pearson Correlation Matrix (Eq. 13)

$$r_{u,v} = \Sigma_d (x\bar{}_{d,u} - x\bar{}_u)(x\bar{}_{d,v} - x\bar{}_v) / [(n-1) \times \sigma_u \times \sigma_v] \quad \textit{(Eq. 13)}$$

All reported correlations are statistical associations only (Corr(X,Y) ≢ C(X→Y)). Causal interpretation requires experimental design outside this framework's scope.

3.7.2 OLS Regression (Eq. 14)

$$\hat{\beta} = (X^T X)^{-1} X^T y \quad \text{where} \quad y = \text{PM10 daily means}, X = \text{meteorological predictors (z-normalised)} \quad \textit{(Eq. 14)}$$

$R^2 = 1 - SS_{res} / SS_{tot}$ measures the fraction of PM10 variance explained by meteorological predictors.

3.8 Grounding Verifier G (Eqs. 15–16) — Novel Operator

Given JSON contract J (a flat numeric lookup table L = {path: value}) and generated narrative N, the grounding property states that every numerical claim c in sentence s must match some j ∈ J within type-adaptive tolerance ε:

$$G(s) \subseteq J \iff \forall c \in C(s) : \exists j \in J : |c-j|/\max(|j|,1) \leq \varepsilon(\text{type}(c)) \quad \textit{(Eq. 15)}$$

The G_rate scalar metric across all numerical sentences N_num:

$$G_rate = |\{s \in N_num : G(s) \subseteq J\}| / |N_num| \in [0, 1] \quad \textit{(Eq. 16)}$$

Type-adaptive tolerance: ε = 0.05 (5% relative) for floating-point concentrations; ε_abs = 2.0 for integer counts; ε_abs = 0.10 for quality scores in [0,1]. G_rate ≥ 0.75 = PASS; G_rate < 0.75 = REVIEW.

3.9 Reproducibility Theorem (Eq. 17)

Theorem (Reproducibility):

For any data state S (the input CSV file) and system version V (ATARS release tag), let J = ATARS(S, V) be the JSON contract produced by the pipeline. Then:

$$\text{SHA-256(canonical_JSON(J)) is invariant under all re-executions of ATARS(S, V)} \quad \textit{(Eq. 17)}$$

Proof: Every formal operator (Eqs. 1–16) is a pure deterministic function of its inputs — no random sampling, no state mutation between calls, and no timestamp-dependent branching. The Isolation Forest uses fixed random_state = 42, which seeds the NumPy random generator before tree construction, ensuring identical tree structures on every execution. The Holt-Winters algorithm uses fixed α, β, γ constants logged in the audit record. The LLM is called at temperature τ = 0 (deterministic output). The canonical JSON serialisation uses sort_keys=True and ensure_ascii=True, guaranteeing identical byte output independent of Python dict insertion order or platform. One potential source of non-determinism that users should be aware of is floating-point arithmetic across different CPU architectures (e.g., x86 vs ARM) — in practice, IEEE 754 compliance ensures identical

results on modern hardware, but cross-architecture reproducibility is not formally guaranteed by the theorem. Within the same hardware architecture and ATARS version, SHA-256(J) is invariant. □

4. System Architecture and Pipeline

ATARS v2.0 is implemented as a single-file Python module (atars_v2.py, 4,449 lines, 44 functions) structured into 13 functional sections. The core entry point run_pipeline() executes a 9-step sequential pipeline with structured console logging, per-step error recovery, and graceful degradation at each step.

4.1 The 9-Step Pipeline

Step	Name	Primary Outputs
1	Data Ingestion	Normalised DataFrame, parsed datetime index, physical plausibility filtering
2	Quality Assessment	Per-record q_i flags, per-day Q(D_v) scores, confidence classification
3	Statistical Analysis	Daily aggregates, rolling baseline B(D,W), z-scores ζ_d, deltas δ_d, bootstrap CIs
3.5	ML Analysis	Isolation Forest anomaly flags, Holt-Winters 14-day PM10 forecast
4	JSON Contract Assembly	Typed data contract J; hash_J = SHA-256(canonical_JSON(J))
5	AI Narrative	LLM-generated section narratives conditioned on J (or statistical placeholders)
5.5	RGV Verification	G_rate ∈ [0,1], per-sentence verdicts, grounding_verification.json
5.5b	Chart Insights	Deterministic data-driven captions for all 15 charts (no LLM)
6	Chart Generation	15 publication-quality figures at 160 DPI
7	Report Compilation	Full 18-section Word (.docx) report with embedded charts
8	Audit Log	audit_log.json with run_id (UUID), hash_J, hash_report, full config snapshot

4.2 Input Specification

The system accepts any CSV or Excel file containing a datetime column and at least one pollutant column. Column names are normalised at load time by a keyword-matching synonym resolver mapping over 40 naming conventions to canonical names. Four encodings are tried in sequence (UTF-8, Latin-1, CP1252, UTF-8-BOM) to handle real-world file encoding issues.

4.3 JSON Contract J — Formal Data Boundary

The JSON contract J is the strictly enforced boundary between the statistical analysis layer and the LLM narrative layer. J contains exclusively aggregated statistics — never raw sensor readings. ML results (Isolation Forest anomaly flags, Holt-Winters forecasts) are explicitly excluded from J. This is a deliberate security decision: if statistical extrapolations were included in J, the LLM could present uncertain predictions as established historical facts.

4.4 Complete Output Package

File	Contents
ATARS_Report_{city}_{date}.docx	18-section analysis report with 15 embedded charts
json_contract.json	Typed data contract J — all inputs to LLM narrative
grounding_verification.json	Per-sentence RGV verdicts and G_rate — reproducible audit artifact
audit_log.json	hash_J (SHA-256), hash_report, run_id (UUID), full config snapshot
charts/chart_01.png – chart_15.png	15 publication-quality figures at 160 DPI
run_atars.bat	Windows Task Scheduler automation batch file

5. Machine Learning Enhancements

ATARS v2.0 adds two ML modules to the core statistical pipeline. Both modules are strictly additive: they do not modify any formal operator (Eqs. 1–16), do not contribute to the JSON contract J, and do not affect the SHA-256 audit hash of the formal layer.

5.1 ML Module 1: Isolation Forest Multivariate Anomaly Detection

5.1.1 Motivation

The formal z-score operator $\zeta_{d,v}$ (Eq. 8) evaluates each variable v independently. A day on which all five major pollutants are simultaneously at 2.5 standard deviations above baseline — a pattern strongly indicative of a regional pollution event — would be missed by univariate z-score analysis because no individual variable exceeds the 3σ threshold. Isolation Forest addresses this by scoring the joint distribution of all pollutant variables simultaneously.

5.1.2 Algorithm

The Isolation Forest [6] builds an ensemble of B = 150 isolation trees. Each tree T_i is constructed by recursively selecting a random feature and a random split point, partitioning data until each instance is isolated. The anomaly score for instance x is:

$$s(x,n) = 2^{-E[h(x)]/c(n)} \quad \text{where} \quad c(n) = 2H(n-1) - 2(n-1)/n \qquad \textit{(Eq. 18)}$$

where E[h(x)] is the expected path length across all trees and c(n) is the normalisation constant. Anomalies have short paths ($s \to 1$); normal points require more splits ($s \to 0$).

5.1.3 ATARS Implementation

The run_isolation_forest() function constructs a feature matrix by pivoting daily_df to an (n_days × 4) array with columns [PM10, NO2, SO2, Ozone] and applying StandardScaler normalisation. IsolationForest is instantiated with n_estimators=150, contamination=0.05, random_state=42. The fixed seed ensures byte-for-byte identical anomaly flags on every re-execution.

5.2 ML Module 2: Holt-Winters Triple Exponential Smoothing

5.2.1 Model Specification

Holt-Winters triple exponential smoothing [9] decomposes the PM10 daily series into level L_t, trend T_t, and additive seasonal component S_t with period m = 7 days:

$$L_t = \alpha(x_t - S_{t-m}) + (1-\alpha)(L_{t-1} + T_{t-1}) \qquad \text{[Level]} \qquad \textit{(Eq. 19)}$$

$$T_t = \beta(L_t - L_{t-1}) + (1-\beta)T_{t-1} \qquad \text{[Trend]} \qquad \textit{(Eq. 20)}$$

$$S_t = \gamma(x_t - L_t) + (1-\gamma)S_{t-m} \qquad \text{[Seasonal]} \qquad \textit{(Eq. 21)}$$

$$\hat{x}_{t+h} = L_t + h \cdot T_t + S_{t-m+((h-1) \bmod m)+1} \qquad \text{[Forecast]} \qquad \textit{(Eq. 22)}$$

The 80% prediction interval (widening with horizon h):

$$PI_{80}(h) = \hat{x}_{t+h} \pm 1.28 \times \sigma_{resid} \times \sqrt{h} \qquad \textit{(Eq. 23)}$$

Default parameters: α=0.30, β=0.10, γ=0.20. The 80% interval rather than 95% reflects the genuine uncertainty of 14-day statistical extrapolation.

5.2.2 Security Design

Forecast values are deliberately excluded from J. This prevents the LLM from encountering statistical extrapolations and potentially stating them as historical measurements. The audit log records RMSE, MAE, α, β, γ, and the 14-day forecast mean for reproducibility without transmitting forecast values to the LLM.

6. Runtime Grounding Verifier (RGV) — Novel Contribution

> **NOVEL CONTRIBUTION STATEMENT** The Runtime Grounding Verifier and the G_rate metric are the primary novel contribution of this paper. To the best of the author's knowledge, this is the first published system that: (1) algorithmically enforces the constraint $G(s) \subseteq J$ for LLM-generated text in an environmental monitoring pipeline; and (2) produces a computable, auditable scalar metric $G_rate \in [0,1]$ quantifying numerical grounding. Prior systems enforced this constraint only via prompt instruction — a probabilistic, unauditable mechanism.

6.1 Motivation

When an LLM generates a report narrative from a data contract J, it may produce sentences containing numerical claims never present in J — confident exceedance counts, correlation coefficients, or anomaly dates that are fabricated. In an environmental monitoring context where erroneous values could influence policy decisions or public health advisories, this failure mode is unacceptable. The RGV addresses this by running post-generation: it receives the complete generated narrative N and the JSON contract J, algorithmically verifies each numerical claim against J, and produces a structured evidence table and G_rate that can be included in the audit log and independently reproduced.

6.2 Comparison with Prior Grounding Systems

Property	FactScore [18]	FaithDial [19]	ATARS RGV
Verification target	Factual atomic claims	Dialogue faithfulness	Numerical values in narrative
Data source required	Web search / retrieval index	Labelled dialogue corpus	JSON contract J (already computed)
Numerical tolerance	No — semantic match only	No — semantic match only	Yes — type-adaptive (±5%, ±2, ±0.1)
Produces audit artifact	No	No	Yes — grounding_verification.json
Runtime	Minutes (retrieval)	Minutes (model inference)	Seconds (pure Python, no network)
Zero-transmission compatible	No — requires web access	No — requires corpus	Yes — fully local
Embedded in pipeline	No — standalone tool	No — standalone tool	Yes — Step 5.5 of ATARS pipeline

6.3 RGV Algorithm

The RGV operates in four sequential stages:

- Stage 1 — J Flattening: The JSON contract J is recursively traversed (depth-first) to produce a flat lookup table L = {dotted.path: float_value}. |L| is logged in the audit record.
- Stage 2 — Claim Extraction: For each sentence s in each narrative section k, numerical claims C(s) are extracted by a regular expression matching integers and floats, including negatives and percentages. Numbers below 30 appearing in isolation are excluded to reduce false positives from section references.
- Stage 3 — Grounding Check: For each claim c ∈ C(s), L is searched for the closest matching value j under:

  $$\text{grounded}(c,j) = |c-j|/\max(|j|,1.0) \leq \varepsilon(\text{type}(c)) \quad \textit{(Eq. 24)}$$
- Stage 4 — G_rate Computation:

  $$G_rate(k) = |\{s \in N_k^{num} : verdict \in \{GROUNDED, PARTIALLY_GROUNDED\}\}| / |N_k^{num}| \quad \textit{(Eq. 25)}$$

  $$G_rate = \Sigma_k\ |N_k^{\{num,grounded\}}| / \Sigma_k\ |N_k^{num}| \quad \textit{(Eq. 26)}$$

6.4 Inherent Limitations of the RGV

- Numerical scope only: The RGV verifies extracted numerical values, not semantic meaning. A sentence stating 'The city air quality is safe' with no numbers passes with G_rate = 1.0 regardless of factual status.
- Semantic gap: A sentence can contain correct numbers interpreted incorrectly and still pass. G_rate is a grounding metric, not a semantic truth detector. Expert review remains essential for sentences that pass G_rate but require domain interpretation.

7. Security Model and Reproducibility

7.1 Zero External Transmission

ATARS v2.0 makes exactly one permitted external network call: POST localhost:11434 (Ollama local LLM server). This call is disabled by the --no-llm flag, in which case no network traffic of any kind is generated. No telemetry, usage analytics, crash reports, or data are transmitted to any external server under any configuration.

7.2 LLM Data Isolation

The LLM receives the JSON contract J only — never raw sensor records. J contains exclusively aggregated statistics with minimum daily granularity. An adversary intercepting the Ollama API call would receive only aggregated annual statistics, not individual measurements or station-identifiable metadata beyond city name.

7.3 ML Security Properties

Both ML modules operate entirely within Python process memory on the daily_df DataFrame. No model weights are persisted to disk. No ML results enter J, preventing transmission to the LLM. Fixed random_state=42 for IF and fixed α,β,γ for HW ensure reproducibility without introducing external dependencies.

7.4 SHA-256 Audit Trail (Eqs. 27–28)

Every ATARS run produces a JSON audit log containing:

```
hash_J = SHA-256(canonical_JSON(J))    [canonical = sort_keys=True, ensure_ascii=True]    (Eq. 27)
hash_report = SHA-256(bytes(ATARS_Report.docx))    (Eq. 28)
```

The canonical JSON serialisation guarantees identical byte output across Python versions and platforms. Any modification to the dataset, configuration parameters, or ATARS version produces a different hash_J, making tampering or parameter drift detectable.

8. Experimental Evaluation and Validation

8.1 Formal Operator Validation (Synthetic)

Each formal operator was validated by cross-checking its output against manual computations on a synthetic 30-day dataset with analytically known values. A daily mean injected exactly 3.0 standard deviations above the rolling baseline was flagged with $|\zeta_d| = 3.0$ (boundary anomaly). A dataset with exactly 90% valid records produced $Q(D) = 0.900$ (HIGH boundary). A perfectly stationary series produced ACF $\tilde{\rho}(k) \approx 0$ for all $k > 0$. All 16 operators produced exact expected outputs.

8.2 Isolation Forest Validation (Synthetic)

The IF module was validated on a 120-day synthetic dataset with 5 variables and 6 intentionally injected multivariate anomalies (all variables simultaneously at 2.5σ). With contamination=0.05 and random_state=42, the system correctly flagged all 6 injected events. The z-score operator, applied independently, flagged 0 of the 6 events — confirming the multivariate added value of IF. The random_state=42 guarantee was verified by running the same dataset 10 times and confirming identical anomaly flag sets on every run.

8.3 Holt-Winters Forecast Validation (Synthetic)

The HW module was validated on a 90-day synthetic series with known weekly seasonality and linear trend. With $\alpha=0.30$, $\beta=0.10$, $\gamma=0.20$, $m=7$, horizon=14, the in-sample RMSE was 10.23 μg/m³ and MAE was 8.20 μg/m³ against the ground-truth series. The 80% prediction interval correctly contained 80.1% of held-out validation points — confirming calibration.

8.4 RGV Validation (Synthetic)

The RGV was validated on a curated set of 50 synthetic sentences: 20 fully grounded, 15 partially grounded, 10 fully ungrounded, and 5 non-numerical. Classification accuracy was 100% for fully grounded, fully ungrounded, and non-numerical sentences. Two partially grounded sentences were misclassified due to rounding at the ε boundary — an inherent limitation of fixed-tolerance matching noted in Section 6.4.

8.5 Pipeline Performance

On a standard development machine (Intel Core i7, 16 GB RAM, Windows 11), the complete 9-step pipeline completes in 2–5 minutes for a typical annual dataset (approximately 61,320 hourly records, 7 variables, 365 days) with --no-llm. With LLM narrative enabled (Ollama, llama3.2 model), add

approximately 5–15 minutes depending on hardware. The Gurugram 2024 case study (8,784 records, 10 variables) completed in approximately 4.2 minutes with LLM enabled.

8.6 Real-World Case Study: Gurugram, Haryana, India (2024)

Dataset: To demonstrate ATARS v2.0 on real environmental data, the system was executed on the Gurugram station data from the publicly available *"Air Quality and Metrological Data in India (2024)"* dataset published on Kaggle by Arnav Tripathi (Apache 2.0 License) [21]. This dataset was compiled from two primary government and open sources: the Central Pollution Control Board (CPCB) Continuous Ambient Air Quality Monitoring Station (CAAQMS) public portal (https://cpcb.nic.in) and Open-Meteo (https://open-meteo.com), the official meteorological data API. Both sources are freely accessible to the public and independently verifiable.

Gurugram (Haryana, India) was selected because: (1) it is one of the most severely air-polluted cities in India, consistently ranking among the top ten globally for PM10 concentrations; (2) it is geographically co-located with the author's institution; and (3) the 2024 dataset provides a full calendar year of hourly data across 10 pollutant and meteorological variables — sufficient to exercise all 28 ATARS equations.

8.6.1 Dataset Summary

City / Station	Gurugram / Complete Gurugram
Date Range	2024-01-01 to 2024-12-31 (full calendar year)
Total Days	366 (2024 is a leap year)
Total Records	8,784 hourly observations
Variables	PM10, NO2, NOx, NH3, SO2, Ozone, Benzene, Toluene, Xylene, MP_Xylene + meteorological
Data Source	CPCB CAAQMS (pollutants) + Open-Meteo (meteorological)
Dataset Citation	Tripathi, A. (2024). Kaggle. Apache 2.0 License. [21]
Report ID	ATARS-20260226-F2C572EC
Pipeline Run	2026-02-26 13:14:30
hash_J (SHA-256)	2cbe1b24cef4ff5bf16c1370941a393d...
ATARS Version	v2.0.0
LLM Model	llama3.2, temperature $\tau = 0$ (deterministic)

8.6.2 Data Quality Results

The overall dataset quality score Q(D) = 0.7442 (MODERATE confidence). Most pollutant variables achieved HIGH confidence (Q ≥ 0.90), with PM10, NO2, Benzene, Toluene, and NH3 all exceeding Q = 0.990. Three variables — CO, Ethylbenzene (Eth_Benzene), and NO — had Q = 0.000, indicating complete absence of measurements at this station for 2024; these variables were excluded from all downstream computation.

Variable	Q(D)	Confidence	Anomaly Days	Exceedance Days
PM10	0.995	HIGH	8	364 (99.5%)
NO2	0.999	HIGH	7	25 (6.8%)
NOx	0.995	HIGH	9	0 (0.0%)
NH3	0.991	HIGH	13	0 (0.0%)
SO2	0.991	HIGH	14	0 (0.0%)
Ozone	0.999	HIGH	12	0 (0.0%)
Benzene	0.997	HIGH	13	366 (100%)
Toluene	0.995	HIGH	19	0 (0.0%)
Xylene	0.923	HIGH	15	0 (0.0%)
CO / Eth_Benzene / NO	0.000	LOW	N/A	N/A

8.6.3 Annual Statistics and Key Findings

Table 3 presents the complete annual statistics for all available pollutant variables. The most striking finding is the severe PM10 contamination: an annual mean of 185.10 μg/m³ represents 4.1× the WHO 24-hour guideline of 45 μg/m³, with 364 of 366 monitored days (99.5%) exceeding the guideline. The peak daily mean of 501.65 μg/m³ occurred in January, consistent with the seasonal pattern identified in the rolling baseline analysis (peak month: January, lowest month: September, σ = 83.80 μg/m³).

Variable	Annual Mean	Std σ	Max	Min	Threshold θ	Exceed Days
PM10	185.10 μg/m³	83.80	501.65	39.69	45.0 μg/m³	364 (99.5%)
NO2	17.00 μg/m³	4.89	32.95	8.96	25.0 μg/m³	25 (6.8%)
NOx	31.60 μg/m³	8.19	67.67	14.71	—	0
NH3	68.56 μg/m³	36.01	204.61	24.90	400.0 μg/m³	0
SO2	9.69 μg/m³	2.69	16.99	2.99	40.0 μg/m³	0
Ozone	31.65 μg/m³	7.31	64.53	18.41	100.0 μg/m³	0

Benzene	4.37 μg/m³	1.04	10.27	2.71	1.7 μg/m³	366 (100%)
Toluene	11.12 μg/m³	4.96	35.24	4.16	—	0

Benzene: The annual mean Benzene concentration of 4.37 μg/m³ is 2.57× the WHO annual reference concentration of 1.7 μg/m³, with exceedances on all 366 monitored days (100%). Benzene is classified as a Group 1 IARC carcinogen. The PM10–humidity Pearson correlation r = −0.363 is consistent with wet deposition (negative association between rainfall and particulate concentration). The strongest inter-pollutant association is NO2–NOx (r = 0.688), reflecting the photochemical relationship between nitrogen species.

8.6.4 OLS Regression Results

OLS regression of PM10 against meteorological predictors (z-normalised) produced $R^2 = 0.4244$ on n = 366 observations, indicating that meteorological variables explain 42.44% of the variance in daily PM10 means. The dominant predictor is atmospheric pressure ($\hat{\beta} = +45.8383$), with humidity showing the strongest negative association ($\hat{\beta} = -41.9372$).

Important: All regression coefficients represent statistical associations only (Eq. 17: Corr(X,Y) ≢ C(X→Y)). These figures do not establish causal meteorological mechanisms and should not be interpreted as such.

8.6.5 Temporal Analysis

PM10 lag-1 autocorrelation $\hat{\rho}(1) = 0.793$ indicates strong temporal memory — today's PM10 concentration is a strong predictor of tomorrow's. This is consistent with the persistence of pollution episodes in Gurugram, where stagnant atmospheric conditions during winter months trap particulate matter for multiple consecutive days. The diurnal PM10 profile peaks at 23:00 (218.3 μg/m³) and reaches its minimum at 16:00 (154.0 μg/m³), with a daily amplitude of 64.4 μg/m³. The near-absence of a weekday–weekend contrast (weekday: 184.8 μg/m³, weekend: 185.7 μg/m³) suggests that the dominant PM10 sources in Gurugram are not primarily traffic-driven but rather reflect regional transport and agricultural burning patterns.

8.6.6 Isolation Forest Results

The Isolation Forest module (150 trees, contamination=5%, random_state=42) operating on the 4-variable feature matrix [PM10, NO2, SO2, Ozone] identified 19 anomaly days (5.2% of 366 days). Of these, 16 days (84.2%) were flagged by IF only — representing multivariate pollution episodes that univariate z-score analysis completely missed. Only 3 days were flagged by both methods (high confidence anomalies), and 5 days were flagged by z-score only (single-variable spikes within multivariate norms).

IF Parameter	Value	Interpretation
Features used	PM10, NO2, SO2, Ozone	4 pollutant variables
n_estimators	150	Ensemble size
contamination	5%	Expected anomaly fraction
random_state	42	Fixed for reproducibility
Total days analysed	366	Full calendar year
Anomaly days (IF)	19 (5.2%)	Days flagged as anomalous
Flagged by BOTH methods	3	High-confidence anomalies
IF-only (missed by z-score)	16 (84.2% of IF flags)	Multivariate episodes — novel detections
Z-score only	5	Univariate spikes, within multivariate norm

8.6.7 Holt-Winters Forecast Results

The Holt-Winters model (α=0.3, β=0.1, γ=0.2, m=7, horizon=14) trained on 366 days of PM10 history produced an in-sample RMSE of 77.89 μg/m³ and MAE of 62.82 μg/m³. The 14-day ahead forecast yielded a mean of 64.5 μg/m³ and a peak of 162.5 μg/m³. The high RMSE reflects the large absolute variability in Gurugram PM10 (σ = 83.80 μg/m³) — a RMSE/σ ratio of 0.93 indicating that the model captures approximately the same magnitude of variation as the underlying series. Per the security design in Section 5.2.2, these forecast values were not transmitted to the LLM.

8.6.8 RGV Grounding Results

The Runtime Grounding Verifier analysed the complete LLM narrative (llama3.2, τ=0) against J (143 lookup entries). The overall G_rate = 92.9% (13/14 numerical sentences grounded) — PASS. Per-section results are presented below:

Section	Total Sent.	Numerical	Grounded	G_rate	Verdict
A — Summary	4	2	2	100.0%	PASS ✓
B — Pollutants	10	6	6	100.0%	PASS ✓
C — Correlations	3	1	1	100.0%	PASS ✓
D — Anomalies	4	2	2	100.0%	PASS ✓
E — Quality	4	2	2	100.0%	PASS ✓
F — Limitations	3	1	0	0.0%	REVIEW ⚠

OVERALL	28	14	13	92.9%	PASS ✓

The single REVIEW section (F_limitations) was triggered because the LLM generated a sentence about result generalizability that referenced specific dates not stored directly in J — a computed value derived from two J fields rather than stored explicitly. This outcome is consistent with the known RGV limitation documented in Section 6.4 and does not represent a hallucinated value.

9. Honest Novelty Assessment and Limitations

9.1 Research Gaps Addressed

ATARS v2.0 was designed to close eight specific gaps in the existing literature. Table 5 provides a transparent assessment of the novelty level for each gap addressed.

Gap	Description	Novelty Level	Justification
G1	End-to-end reproducible AQ pipeline	MODERATE	Prior art: openair [3] — ATARS adds formal ops, LLM, RGV
G2	Algorithmic LLM grounding verification	HIGH	No published precedent for closed-contract numerical G_rate
G3	Multivariate + univariate anomaly comparison	MODERATE	IF known; ATARS adds formal IF vs ζ_d comparison framework
G4	Holt-Winters 14-day AQ forecast	LOW	HW known; implementation is engineering contribution
G5	Equation-numbered formal operators	MODERATE	Some prior formalisation exists; ATARS is more complete
G6	JSON contract as LLM data boundary	MODERATE	RAG literature [14] precedes this; AQ context is novel
G7	SHA-256 audit trail for AQ reports	MODERATE	Audit trails known; application to AQ pipeline is novel
G8	Deterministic per-chart LLM-free insights	LOW	Template-based generation — engineering value only

Only Gap G2 (algorithmic LLM grounding verification with G_rate) is assessed as HIGH novelty. Gaps G1, G3, G5, G6, and G7 are MODERATE: ATARS addresses each cleanly, but prior art exists for each component individually. Gaps G4 and G8 are LOW novelty engineering contributions with clear practical value.

9.2 Documented System Limitations

Limitation	Severity	Impact
Fixed HW smoothing parameters (α, β, γ)	MODERATE	Suboptimal forecast on non-standard seasonality
RGV verifies numbers only — not semantics	MODERATE	Semantically wrong sentences with correct numbers pass

IF contamination fixed at 5%	LOW	Always flags ~5% of days regardless of true prevalence
OLS assumes linear meteorological relationships	MODERATE	Non-linear interactions not captured
Single-station design — no spatial interpolation	LOW	Results not transferable across stations without re-run
Cross-architecture floating point variance	LOW	SHA-256 reproducibility limited to same hardware arch

9.3 Future Research Directions

- Replacing fixed HW parameters with Bayesian optimisation or grid-search cross-validation, with parameter sets logged in the audit trail.
- Extending the RGV to verify semantic claims (not just numerical values) using a closed-domain natural language inference model trained on AQ domain text.
- Multi-station extension with spatial interpolation (kriging or IDW) for network-level analysis across CPCB CAAQMS cities.
- ARIMA/SARIMA modelling with auto-order selection (auto-ARIMA) as an alternative or complement to Holt-Winters.
- Integration of a second LLM for cross-model narrative consistency checking — comparing G_rate across two independently generated reports from different models.

10. Conclusion

This paper has presented ATARS v2.0, a complete, reproducible, formally specified framework for automated urban air quality analysis. The system defines 16 equation-numbered core statistical operators (Eqs. 1–16) covering the full analysis lifecycle from data quality assessment through anomaly detection, temporal dependency analysis, multivariate correlation, and OLS regression — supplemented by 12 additional equations for the ML modules, the grounding verifier, and the SHA-256 reproducibility audit (Eqs. 17–28).

Two machine learning modules — Isolation Forest multivariate anomaly detection and Holt-Winters triple exponential smoothing — extend the system's capabilities as strictly additive components, deliberately excluded from the JSON data contract J to prevent LLM generation of uncertain ML extrapolations as established historical facts.

The primary contribution is the **Runtime Grounding Verifier (RGV)** and the G_rate metric (Eqs. 15–16, 24–26). To the best of the author's knowledge, this is the first published system to algorithmically enforce and measure the grounding constraint $G(s) \subseteq J$ for LLM-generated text in an environmental monitoring pipeline, producing a computable, auditable scalar metric that can be independently reproduced from the grounding_verification.json artifact.

The system was validated on both synthetic datasets with analytically known ground truth and a real-world case study on Gurugram, Haryana, India (January–December 2024) using publicly available CPCB data (8,784 records, Apache 2.0 license [21]). The real-world demonstration achieved G_rate = 92.9% (PASS), confirming that the LLM narrative was numerically grounded in the data contract for 13 of 14 numerical sentences. The Isolation Forest identified 16 multivariate anomaly days that z-score analysis missed entirely, demonstrating the concrete diagnostic value of the ML layer.

The Gurugram 2024 results reveal a severe air quality crisis: PM10 exceeded WHO guidelines on 99.5% of monitored days (annual mean 185.10 μg/m³ vs threshold 45.0 μg/m³), and Benzene — a Group 1 IARC carcinogen — exceeded its WHO annual reference concentration on every single day of the year (annual mean 4.37 μg/m³ vs reference 1.7 μg/m³). These findings underscore the public health importance of the reproducible, auditable reporting infrastructure that ATARS provides.

Full source code (atars_v2.py, 4,449 lines, 44 functions) is available at the author's GitHub repository. The Gurugram 2024 run can be exactly reproduced using the public Kaggle dataset [21] and verifying SHA-256(J) = 2cbe1b24cef4ff5bf16c1370941a393d...

"A framework built for researchers, by a researcher. Every number verifiable. Every claim auditable."

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